MDS5202 Assignment 2

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Conceptual

1. Describe the null hypotheses to which the p-values given in Table 1 correspond. Explain what conclusions you can draw based on these p-values. Your explanation should be phrased in terms of *sales, TV, radio, and newspaper*, rather than in terms of the coefficients of the linear model.

Table 1: For the Advertising data, least squares coefficient estimates of the multiple linear regression of number of units sold on TV, radio, and newspaper advertising budgets.

	Coefficient	**Std. error**	**t-statistic**	**p-value**
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

5. Consider the fitted values that result from performing linear regression without an intercept. In this setting, the ith fitted value takes the form $\hat{y}_i = x_i\hat{\beta}$, where

$$\hat{\beta} = \left(\sum_{i=1}^{n} x_i y_i\right) \Bigg/ \left(\sum_{i'=1}^{n} x_{i'}^2\right).$$

Show that we can write $\hat{y}_i = \sum_{i'=1}^{n} a_{i'} y_{i'}$. What is $a_{i'}$?

Note: We interpret this result by saying that the fitted values from linear regression are linear combinations of the response values.

Applied

Data Loading Instructions:
To complete the following questions, please ensure you have the `ISLR` package installed and load the `Auto` dataset using the following R code:

```
# Load the required library and data
install.packages("ISLR") # Execute once if needed
library(ISLR)
data(Auto)
head(Auto)
Auto <- na.omit(Auto) # Recommended:  remove missing values
```

8. This question involves the use of simple linear regression on the `Auto` data set.

(a) Use the `lm()` function to perform a simple linear regression with `mpg` as the response and `horsepower` as the predictor. Use the `summary()` function to print the results. Comment on the output. For example:

i. Is there a relationship between the predictor and the response?

ii. How strong is the relationship between the predictor and the response?

iii. Is the relationship between the predictor and the response positive or negative?

iv. What is the predicted `mpg` associated with a `horsepower` of 98? What are the associated 95% confidence and prediction intervals?

(b) Plot the response and the predictor. Use the `abline()` function to display the least squares regression line.

(c) Use the `plot()` function to produce diagnostic plots of the least squares regression fit. Comment on any problems you see with the fit.

9. This question involves the use of multiple linear regression on the `Auto` data set.

(a) Produce a scatterplot matrix which includes all of the variables in the data set using `pairs()`.

(b) Compute the matrix of correlations between the variables using the function `cor()`. You will need to exclude the `name` variable, which is qualitative.

(c) Use the `lm()` function to perform a multiple linear regression with `mpg` as the response and all other variables except `name` as the predictors. Use the `summary()` function to print the results.

(d) Use the `plot()` function to produce diagnostic plots of the linear regression fit. Comment on any problems you see with the fit (outliers, high leverage points, etc.).

(e) Use the `*` and `:` symbols to fit linear regression models with interaction effects. Do any interactions appear to be statistically significant?

(f) Try a few different transformations of the variables, such as $\log(X)$, $\sqrt{X}$, X^2. Comment on your findings.